

Sample Spaces

math-foundations / 17-sample-spaces

First, an example

Before any abstraction, picture a single roll of a fair six-sided die. The *sample space* — the set of every outcome that could occur — is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Read aloud: Omega equals the set containing 1, 2, 3, 4, 5, and 6. The event “the roll is even” is the subset

$$A = \{2, 4, 6\} \subseteq \Omega.$$

Read aloud: A equals the set containing 2, 4, and 6, which is a subset of Omega. Under the uniform probability measure, $P(A) = 3/6 = 1/2$. Everything that follows generalises this picture.

1. Introduction

Probability theory begins with a single primitive: a set whose elements are the possible outcomes of an experiment. This set is the *sample space*. Every subsequent concept — events, random variables, expectations, conditional distributions — is built on top of this primitive. Before we can assign probabilities, we must first agree on what we are assigning probabilities *to*.

2. Definitions

Definition 1 (Sample Space). A *sample space* is a non-empty set Ω whose elements represent the mutually exclusive and collectively exhaustive outcomes of an experiment or random phenomenon.

Definition 2 (Outcome). An *outcome* is an element $\omega \in \Omega$. It represents one complete atomic realization of the experiment.

Definition 3 (Event). An *event* is a subset $A \subseteq \Omega$. We say the event A *occurs* on outcome ω if $\omega \in A$. In the finite case, the family of all events is the power set

$$2^\Omega = \{A : A \subseteq \Omega\}.$$

Read aloud: Two to the Omega equals the set of all A such that A is a subset of Omega.

Example 4 (Coin flip). $\Omega = \{H, T\}$. The events are $\emptyset, \{H\}, \{T\}, \{H, T\}$.

Example 5 (Die roll). $\Omega = \{1, 2, 3, 4, 5, 6\}$. The event “even” is $\{2, 4, 6\}$.

Example 6 (Two dice). $\Omega = \{1, \dots, 6\}^2$, with $|\Omega| = 36$. The event “sum is 7” is $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$.

Example 7 (Infinite coin sequences). $\Omega = \{H, T\}^\mathbb{N}$, an uncountable set. This is the natural sample space for asymptotic statements such as the strong law of large numbers.

3. The Power Set Theorem

Theorem 8 (Power Set Cardinality). *If Ω is a finite set with $|\Omega| = n$, then the power set 2^Ω has cardinality 2^n .*

Proof. We proceed by induction on n .

Base case ($n = 0$). The unique set of cardinality 0 is $\Omega = \emptyset$. Its only subset is $\emptyset$ itself, so $|2^\emptyset| = 1 = 2^0$.

Inductive step. Assume the claim holds for all sets of cardinality n . Let Ω have cardinality $n + 1$. Fix any element $x \in \Omega$ and let $\Omega' = \Omega \setminus \{x\}$, which has cardinality n .

Partition the subsets of Ω into two classes:

1. Subsets that do *not* contain x — these are exactly the subsets of Ω' , and by the inductive hypothesis there are 2^n of them.
2. Subsets that *do* contain x — each such subset has the form $A \cup \{x\}$ where $A \subseteq \Omega'$. The map $A \mapsto A \cup \{x\}$ is a bijection between $2^{\Omega'}$ and $\{B \subseteq \Omega : x \in B\}$, so there are again 2^n such subsets.

The two classes are disjoint and exhaust 2^Ω , so

$$|2^\Omega| = 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}.$$

Read aloud: The cardinality of two to the Omega equals two to the n plus two to the n, which equals two times two to the n, which equals two to the n plus one. This completes the induction. $\square$

Corollary 9. *A coin flip ($n = 2$) admits $2^2 = 4$ events; a die ($n = 6$) admits $2^6 = 64$ events; two dice ($n = 36$) admit $2^{36} \approx 6.87 \times 10^{10}$ events.*

4. Remarks on the Infinite Case

When Ω is uncountable (e.g. $\Omega = [0, 1]$), one cannot in general use the full power set 2^Ω as the event family — there exist subsets to which no translation-invariant probability measure can be consistently assigned (the Vitali construction). Instead one restricts to a σ -algebra $\mathcal{F} \subseteq 2^\Omega$, typically the Borel σ -algebra. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is then called a *probability space*; sample spaces are the first slot of this triple. See node 18-sigma-algebras for the continuation.